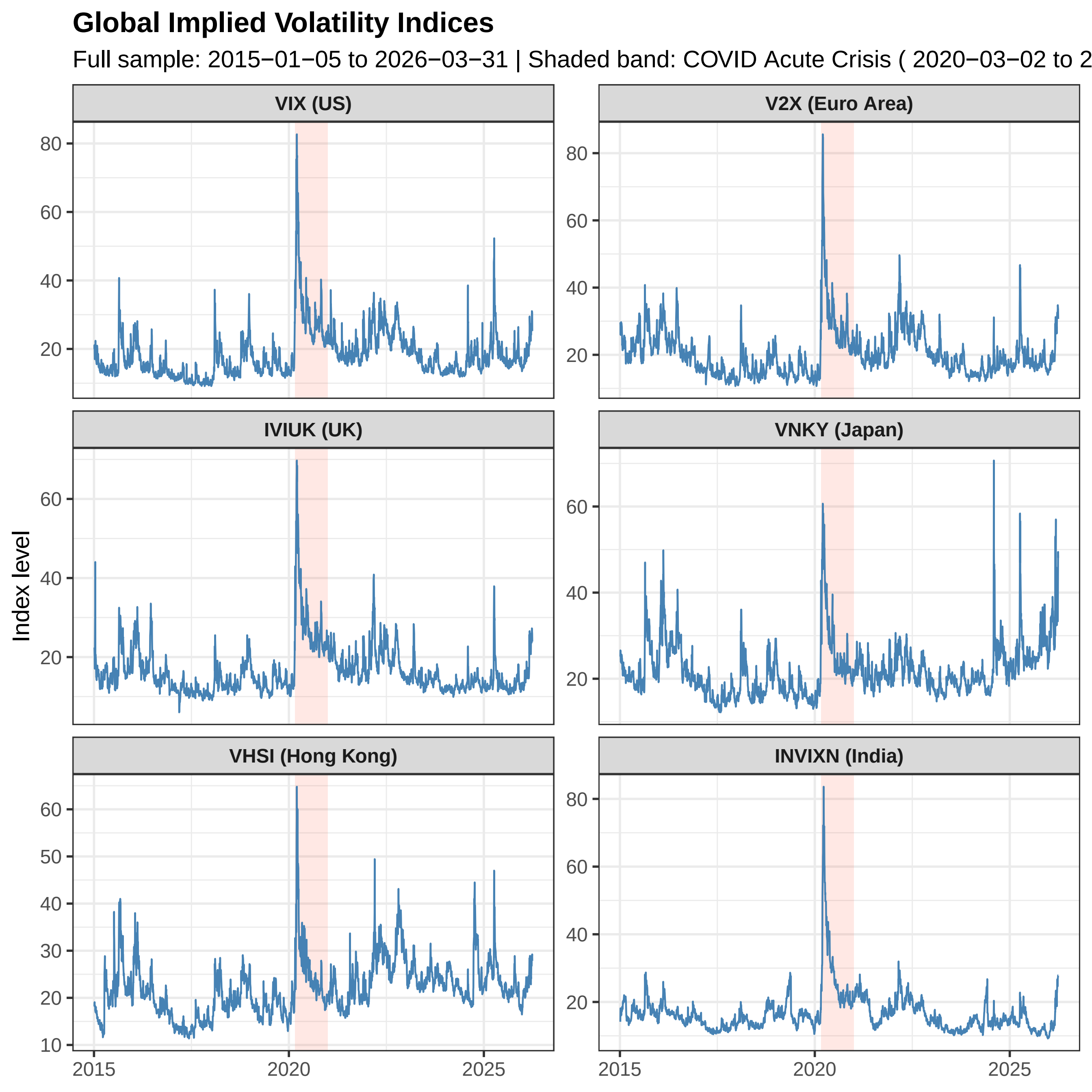
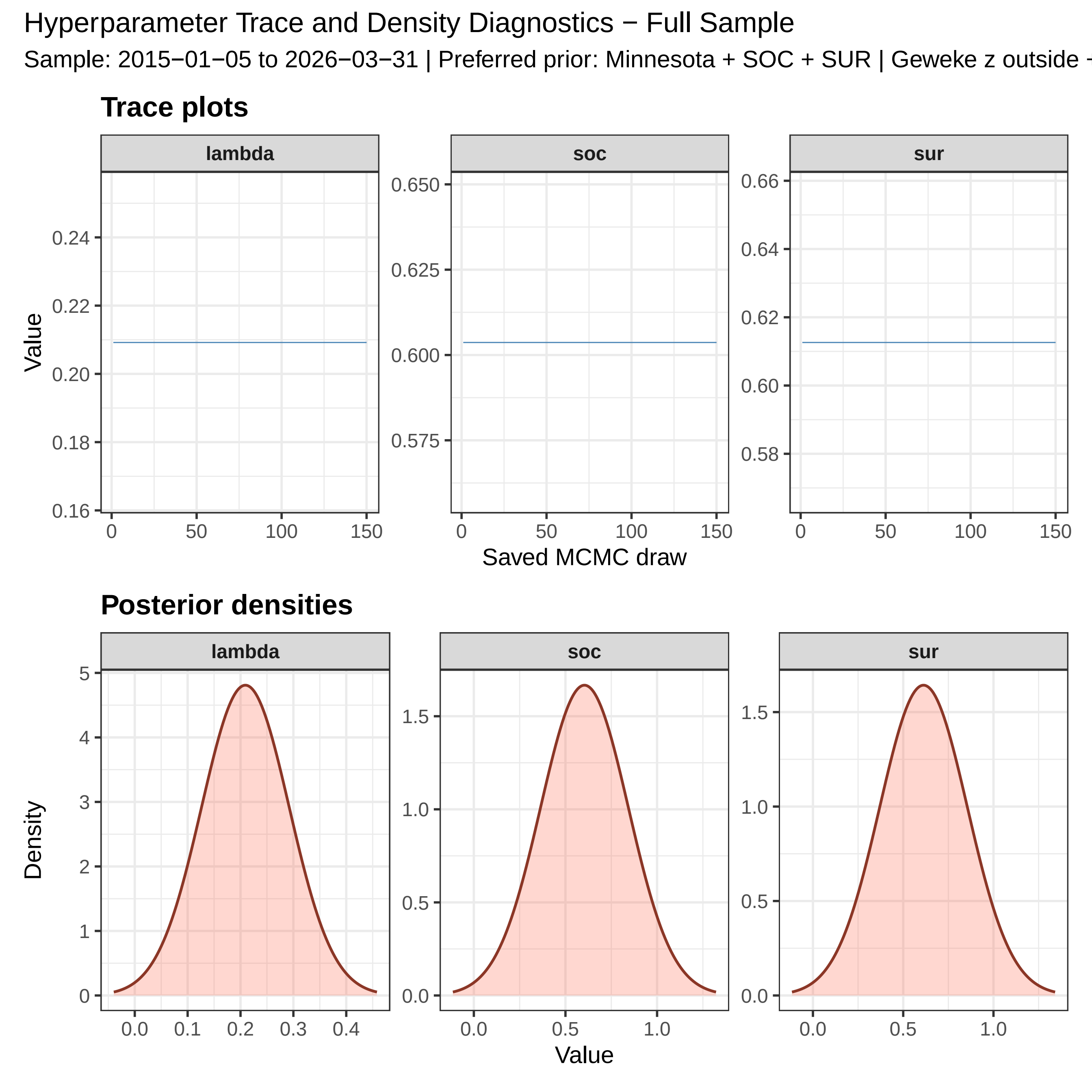
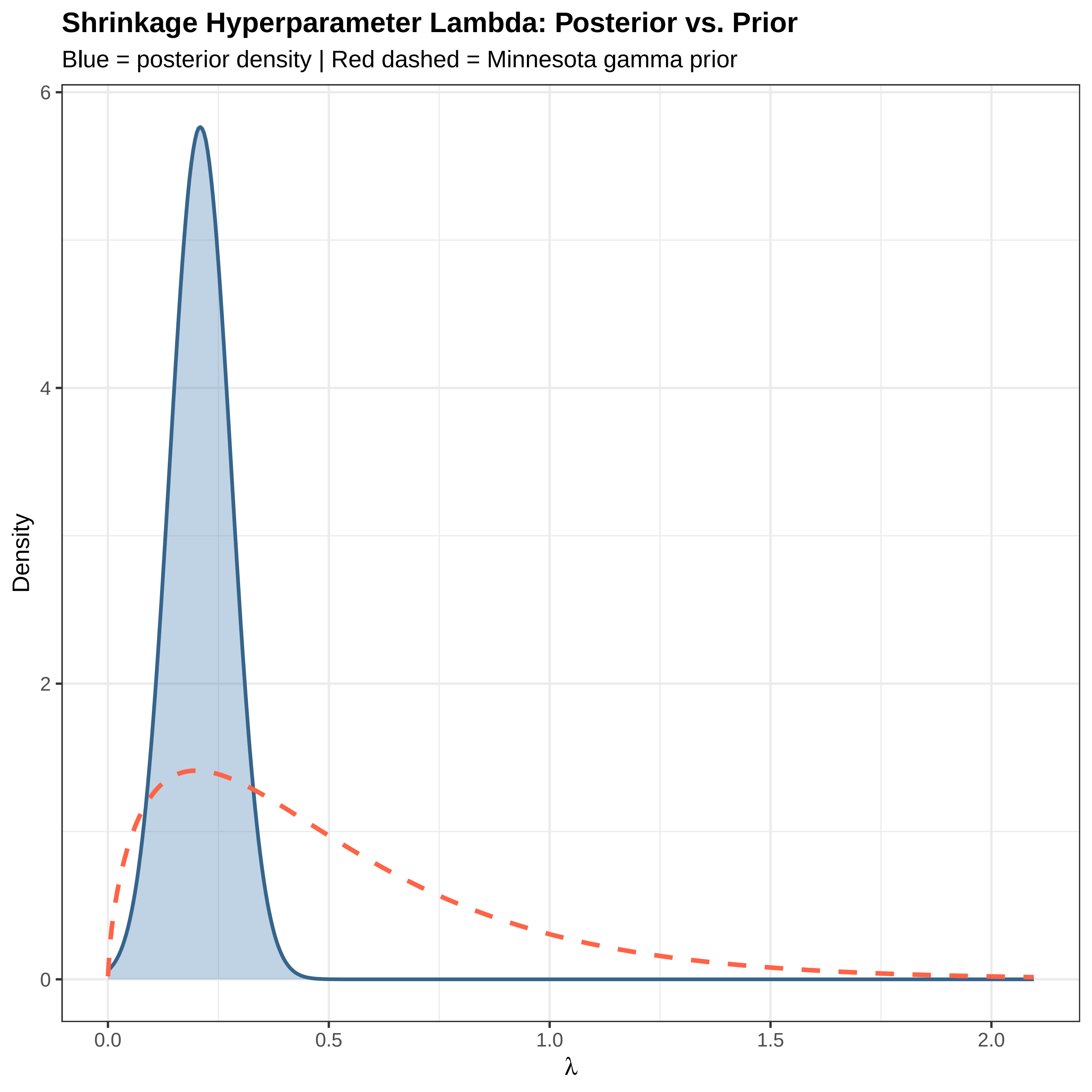
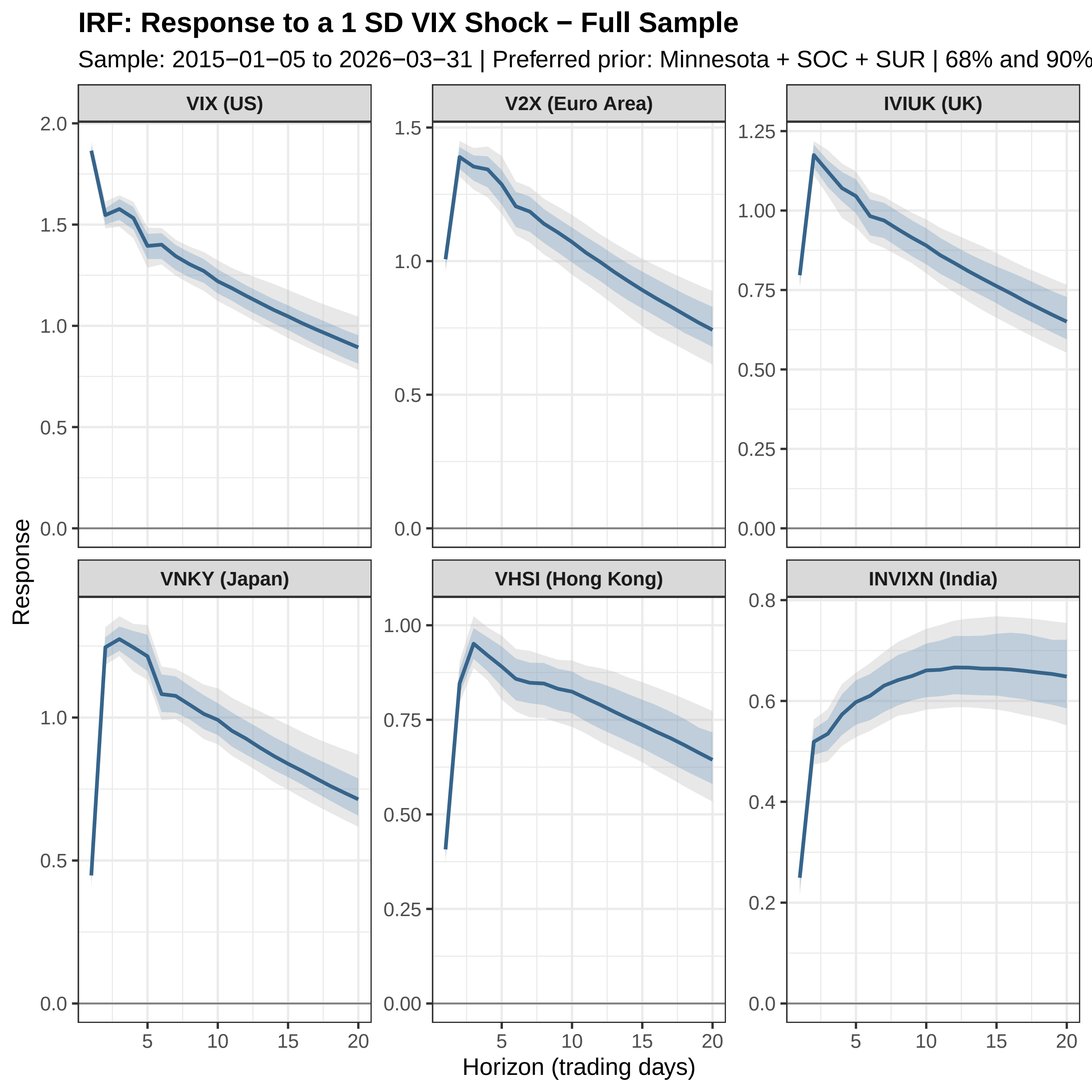


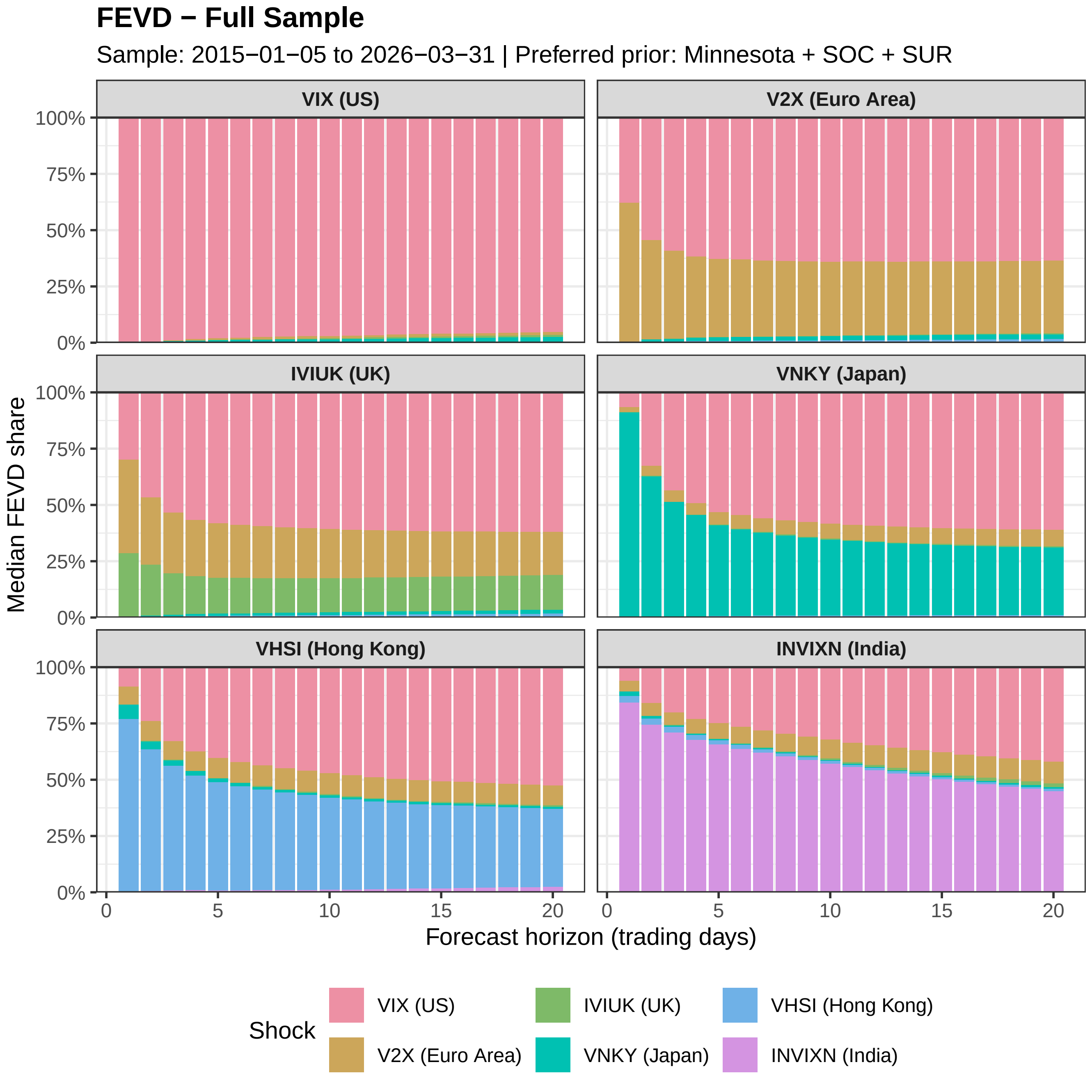
Global Implied Volatility Indices
Full sample: 2015−01−05 to 2026−03−31 | Shaded band: COVID Acute Crisis (2020−03−02 to 2
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
2015
2020
2025

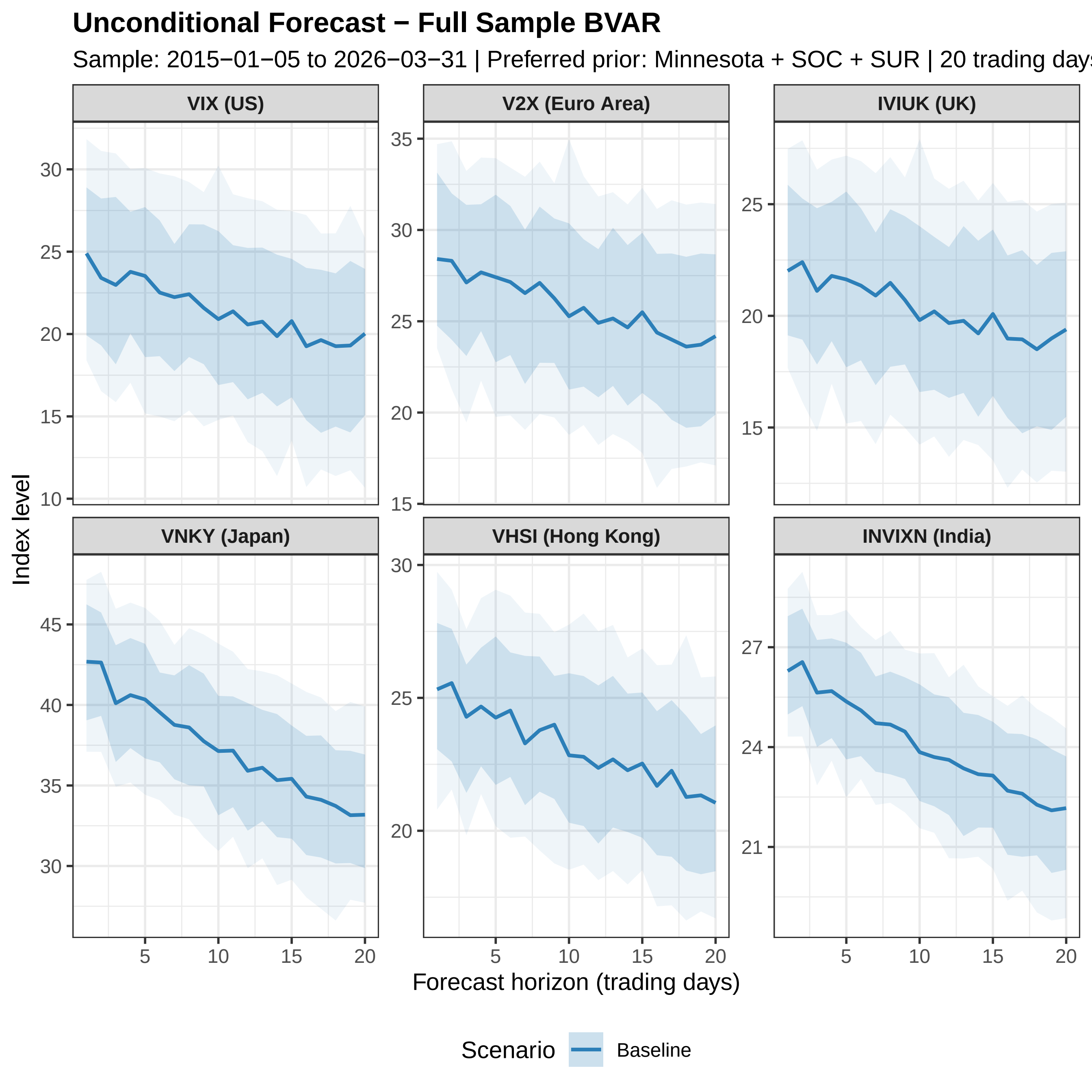

Hyperparameter Trace and Density Diagnostics − Full Sample
Sample: 2015−01−05 to 2026−03−31 | Preferred prior: Minnesota + SOC + SUR | Geweke z outside
Trace plots
lambda
soc
sur
Value
0.16
0.18
0.20
0.22
0.24
0.575
0.600
0.625
0.650
0.58
0.60
0.62
0.64
0.66
0
50
100
150
Saved MCMC draw
Posterior densities
Density
0
1
2
3
4
5
0.0
0.5
1.0
1.5
0.0
0.1
0.2
0.3
0.4
Value


Shrinkage Hyperparameter Lambda: Posterior vs. Prior
Blue = posterior density | Red dashed = Minnesota gamma prior
Density
6
4
2
0
0.0
0.5
1.0
1.5
2.0
λ

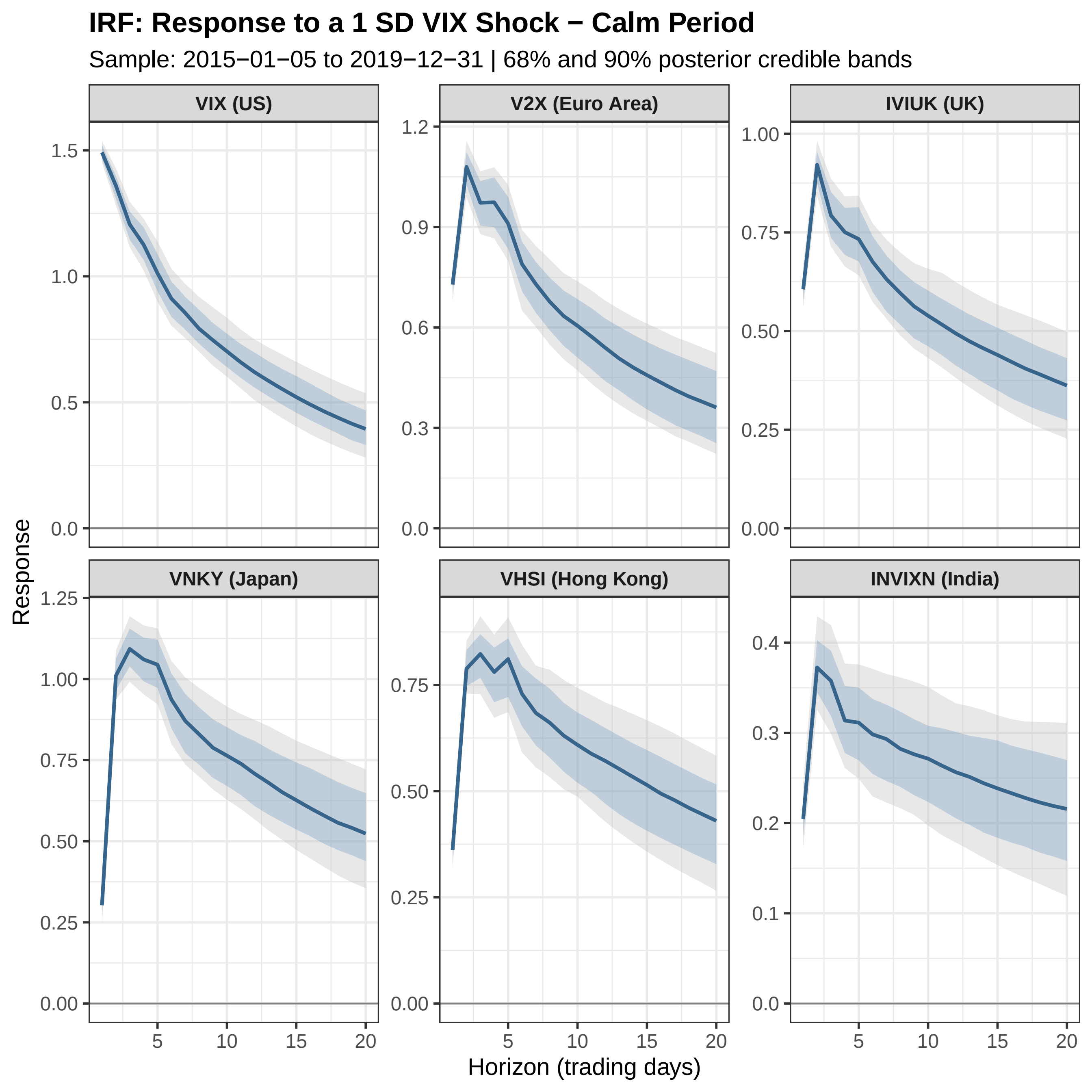


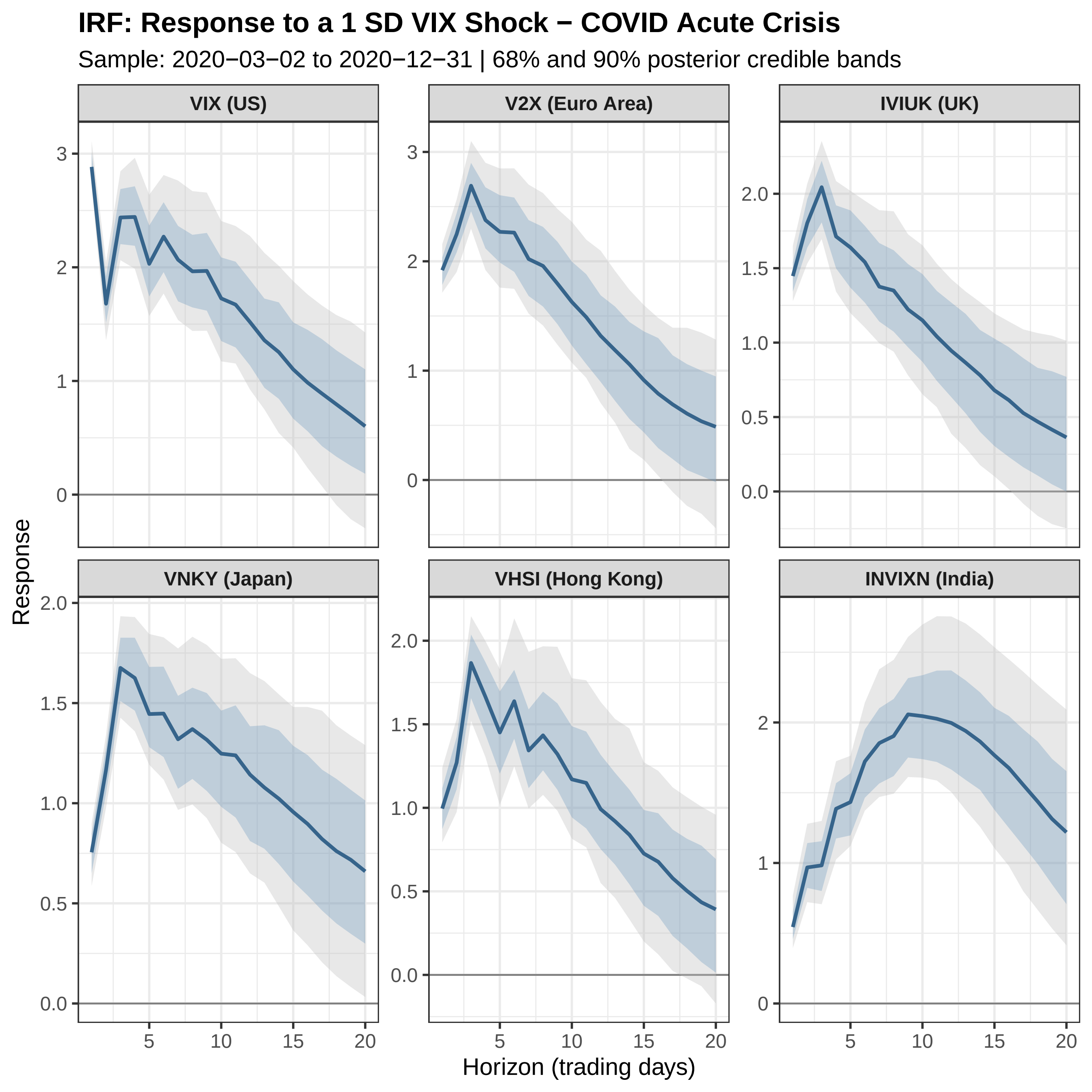
IRF: Response to a 1 SD VIX Shock − Full Sample
Sample: 2015−01−05 to 2026−03−31 | Preferred prior: Minnesota + SOC + SUR | 68% and 90%
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)
0.0
0.5
1.0
1.5
2.0
0.00
0.25
0.50
0.75
1.00
1.25
0.2
0.4
0.6
0.8
5
10
15
20

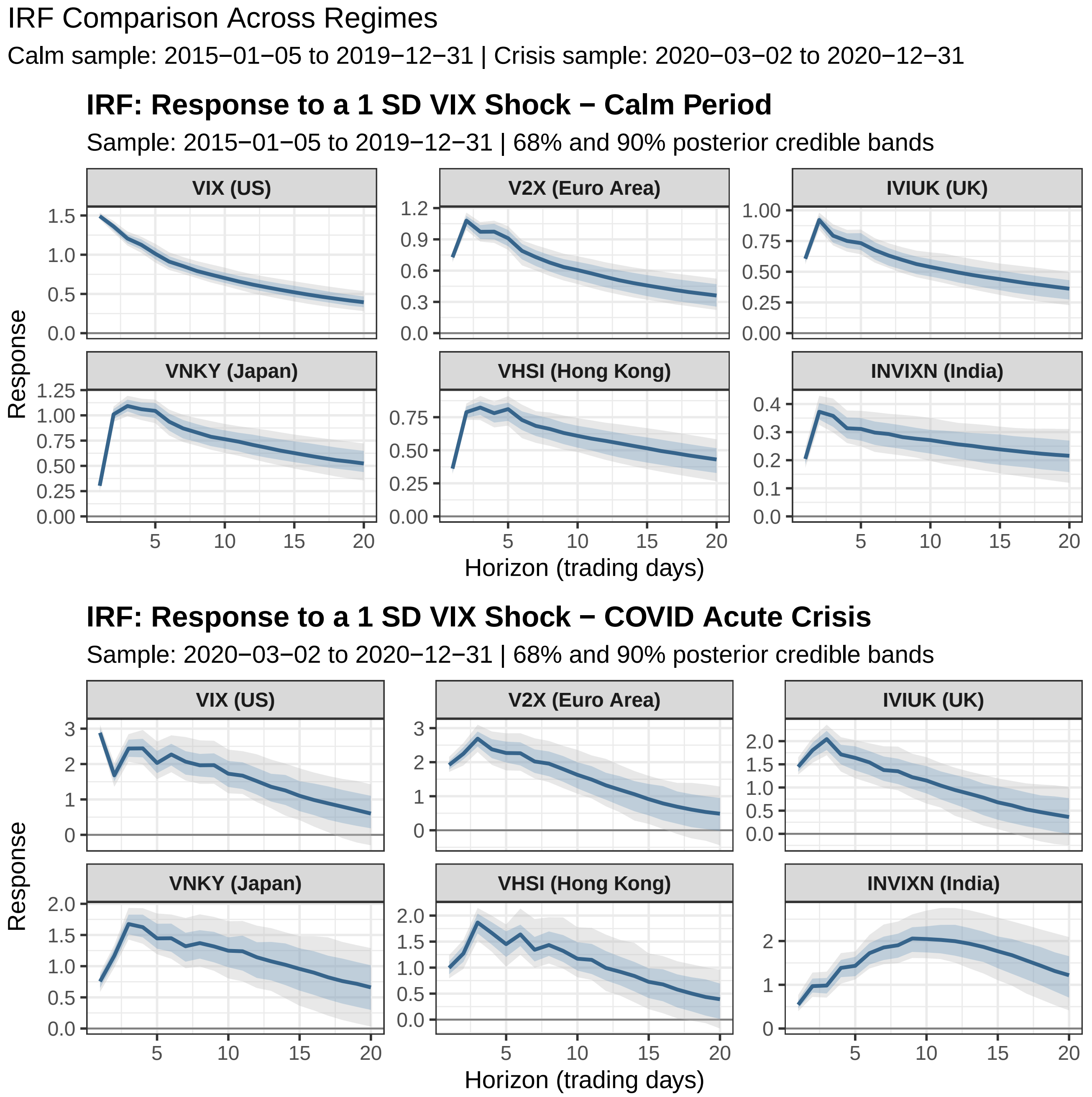

FEVD − Full Sample
Sample: 2015−01−05 to 2026−03−31 | Preferred prior: Minnesota + SOC + SUR
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Median FEVD share
100%
75%
50%
25%
0%
0
5
10
15
20
Forecast horizon (trading days)
Shock
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)

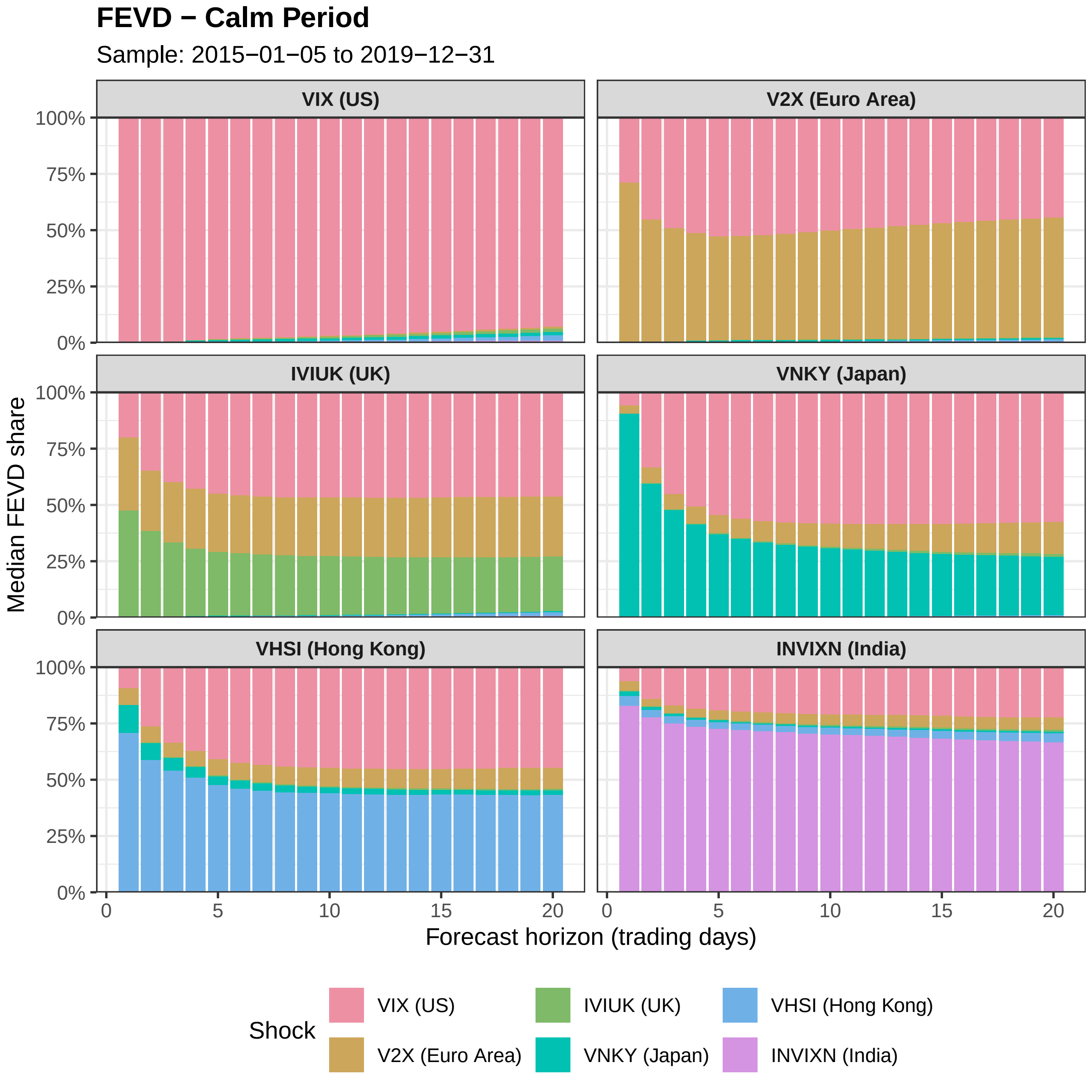


Unconditional Forecast − Full Sample BVAR
Sample: 2015−01−05 to 2026−03−31 | Preferred prior: Minnesota + SOC + SUR | 20 trading days
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline

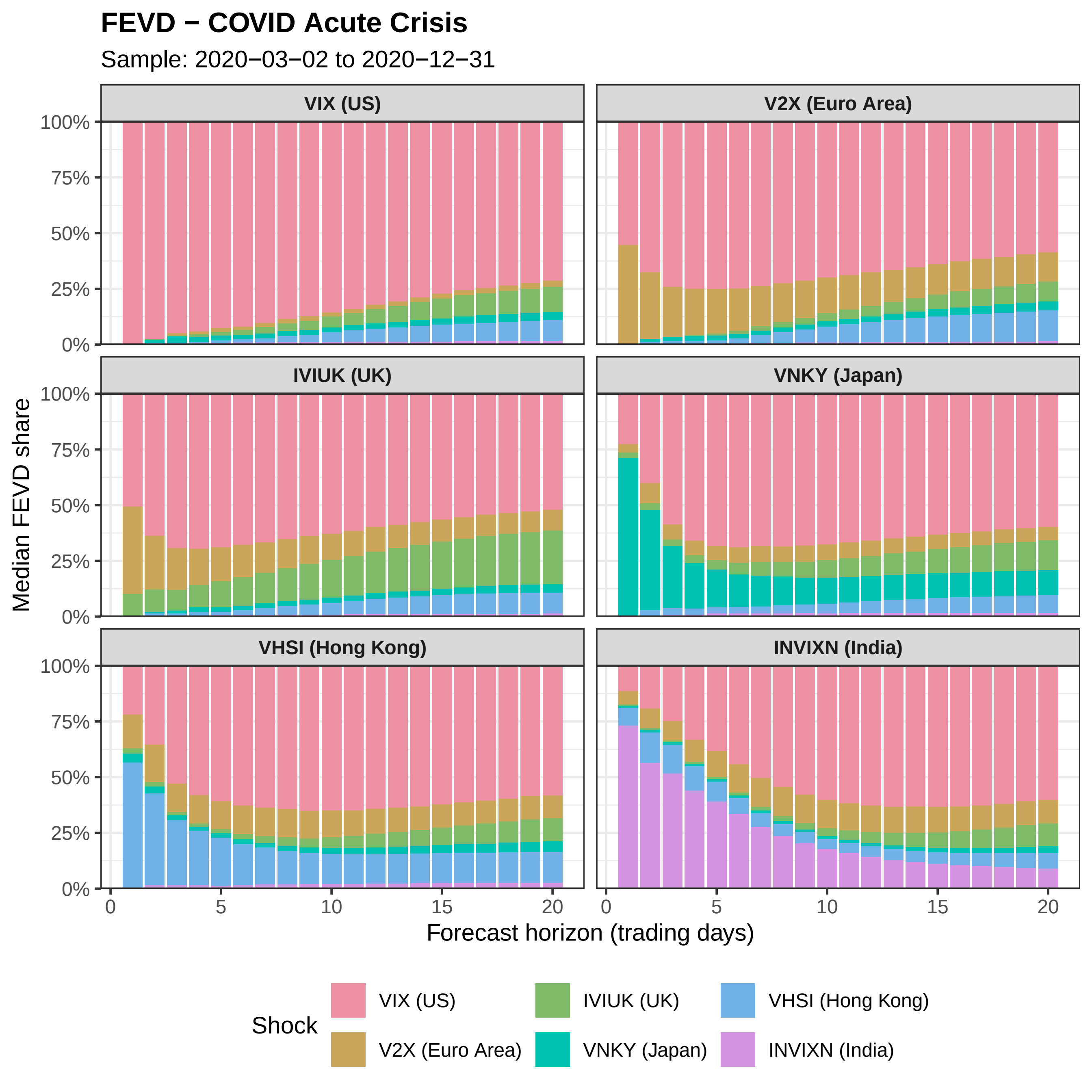

IRF: Response to a 1 SD VIX Shock − Calm Period
Sample: 2015−01−05 to 2019−12−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)

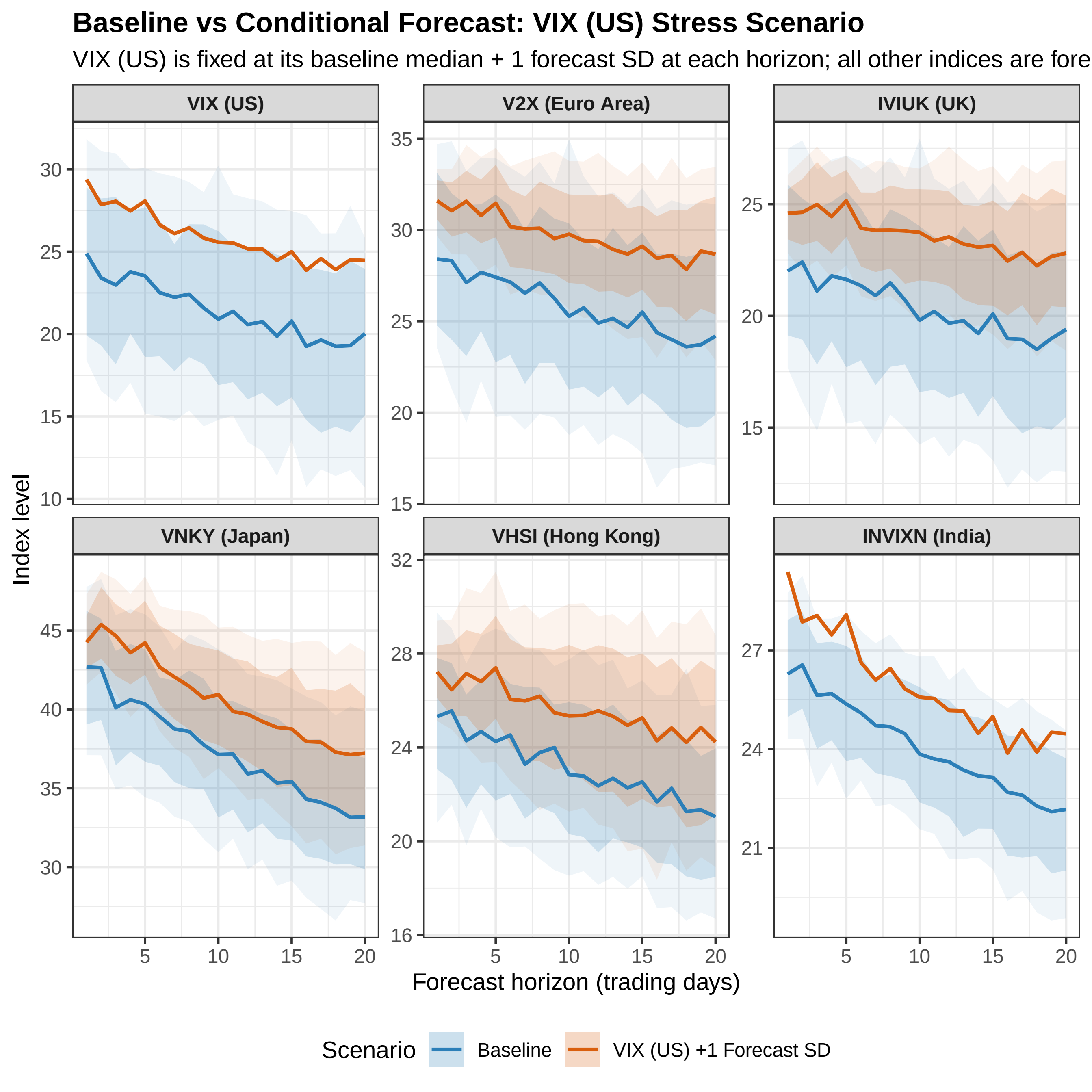

IRF: Response to a 1 SD VIX Shock − COVID Acute Crisis
Sample: 2020−03−02 to 2020−12−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)

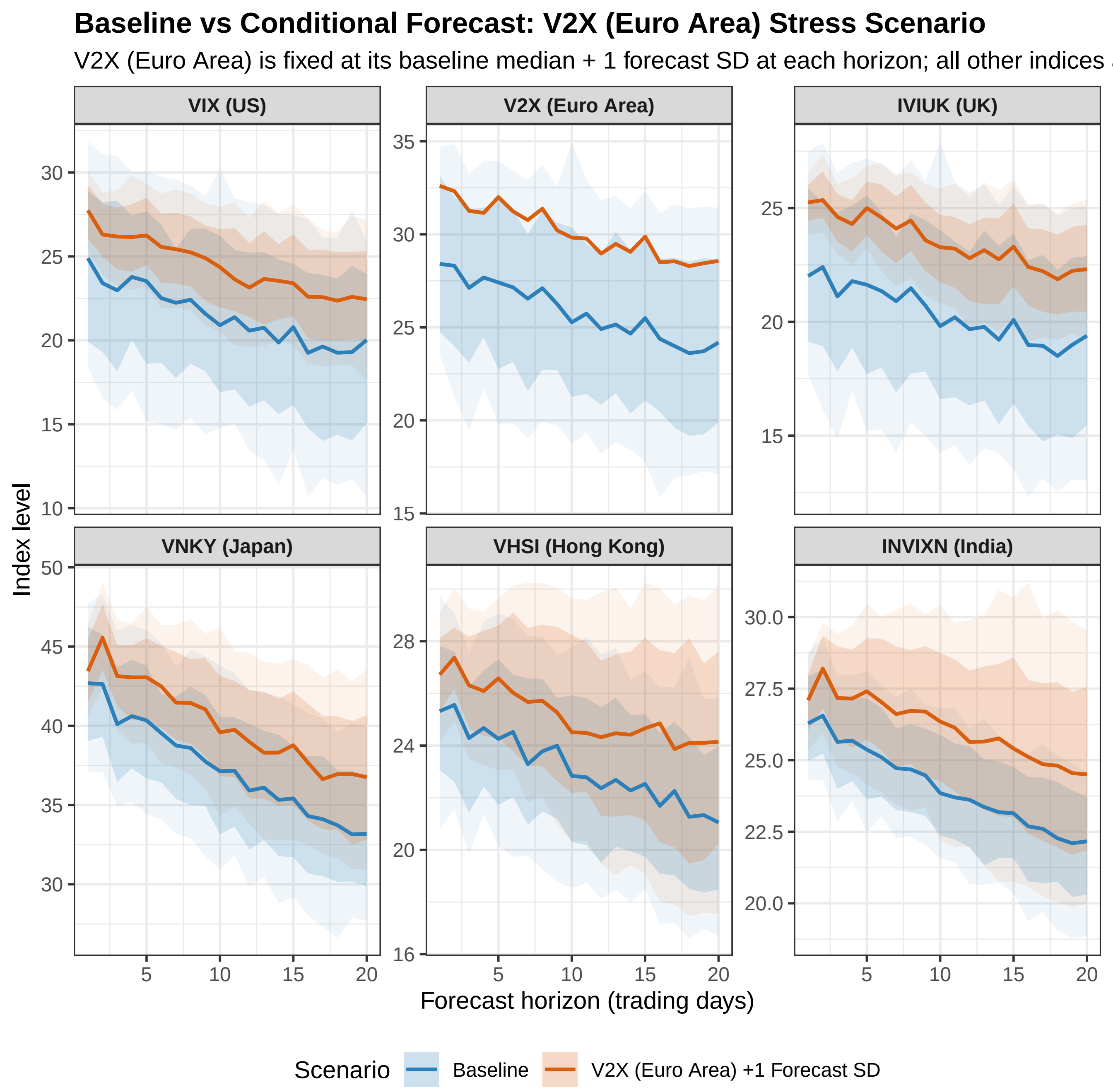

IRF Comparison Across Regimes
Calm sample: 2015−01−05 to 2019−12−31 | Crisis sample: 2020−03−02 to 2020−12−31
IRF: Response to a 1 SD VIX Shock − Calm Period
Sample: 2015−01−05 to 2019−12−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)
IRF: Response to a 1 SD VIX Shock − COVID Acute Crisis
Sample: 2020−03−02 to 2020−12−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)

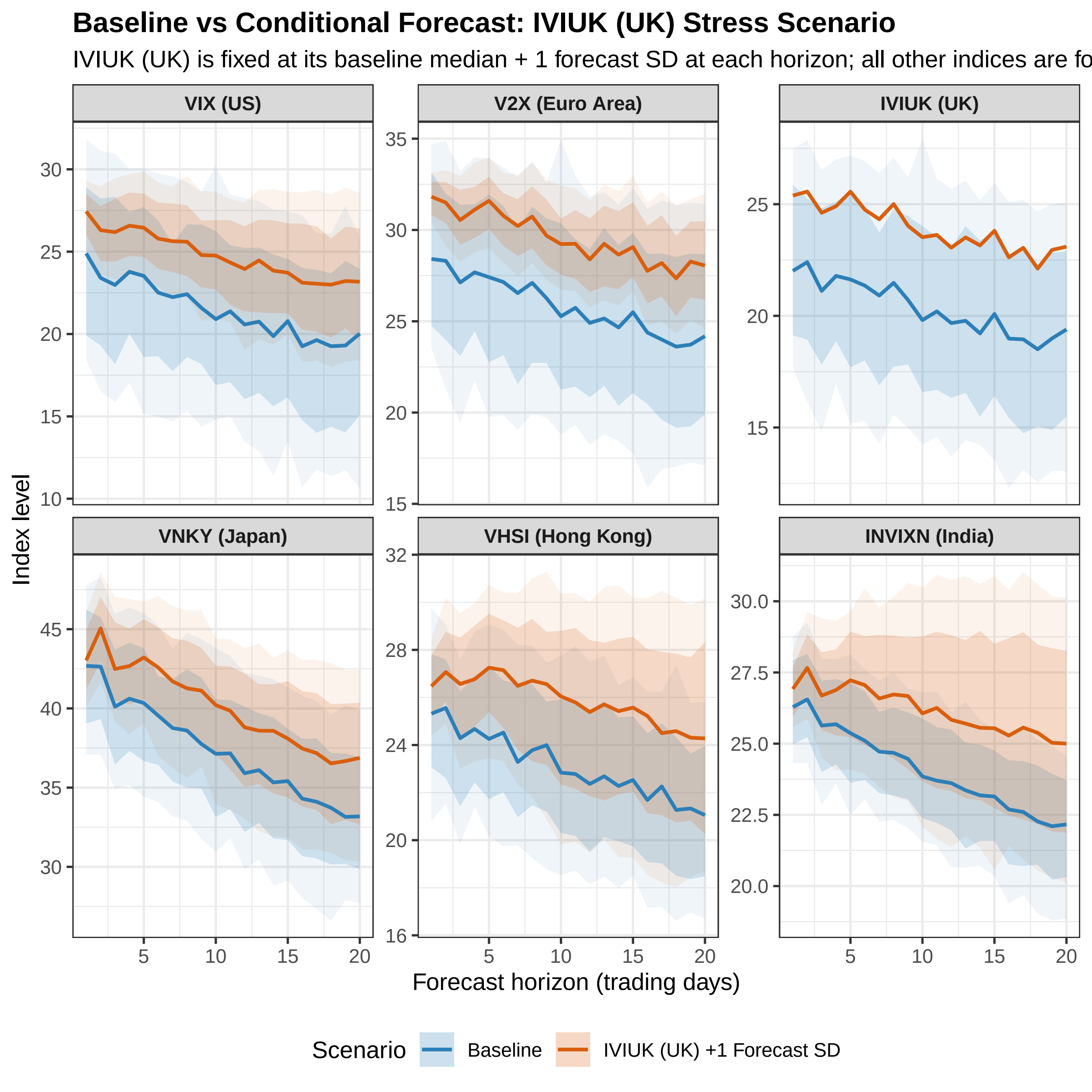


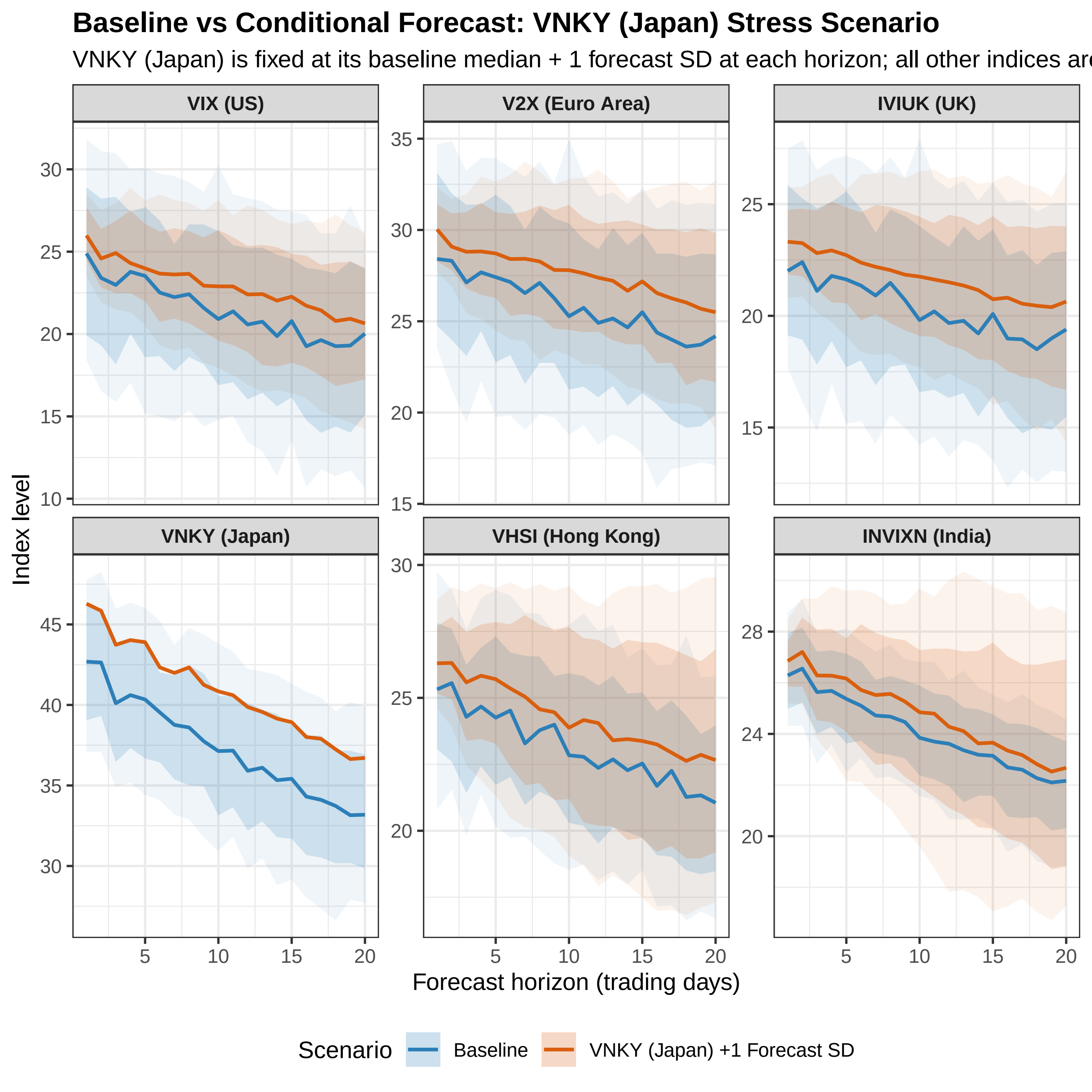
FEVD − Calm Period
Sample: 2015−01−05 to 2019−12−31
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Median FEVD share
100%
75%
50%
25%
0%
0
5
10
15
20
Forecast horizon (trading days)
Shock
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)


FEVD − COVID Acute Crisis
Sample: 2020−03−02 to 2020−12−31
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Median FEVD share
100%
75%
50%
25%
0%
0
5
10
15
20
Forecast horizon (trading days)
Shock
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)

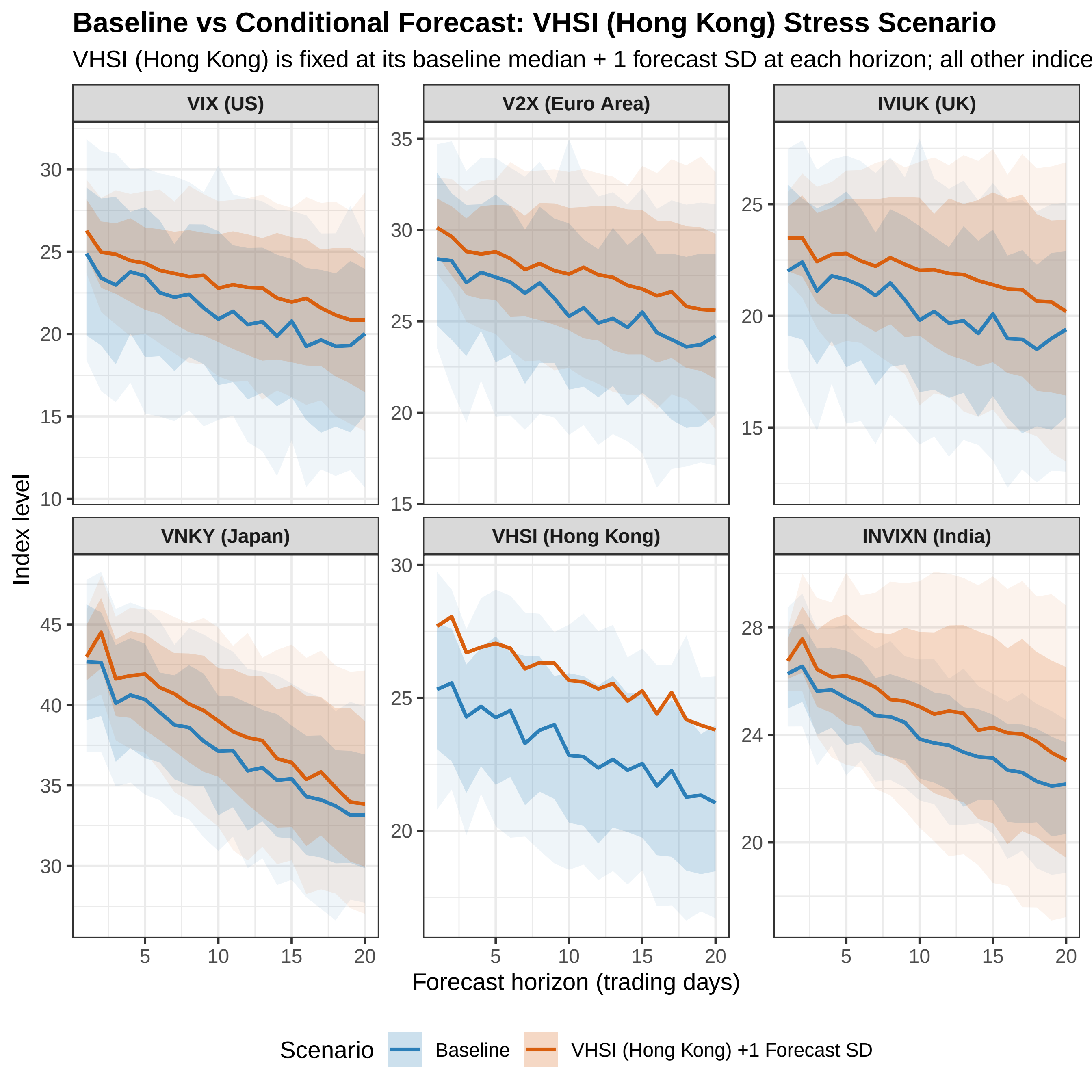

Baseline vs Conditional Forecast: VIX (US) Stress Scenario
VIX (US) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are fore
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VIX (US) +1 Forecast SD

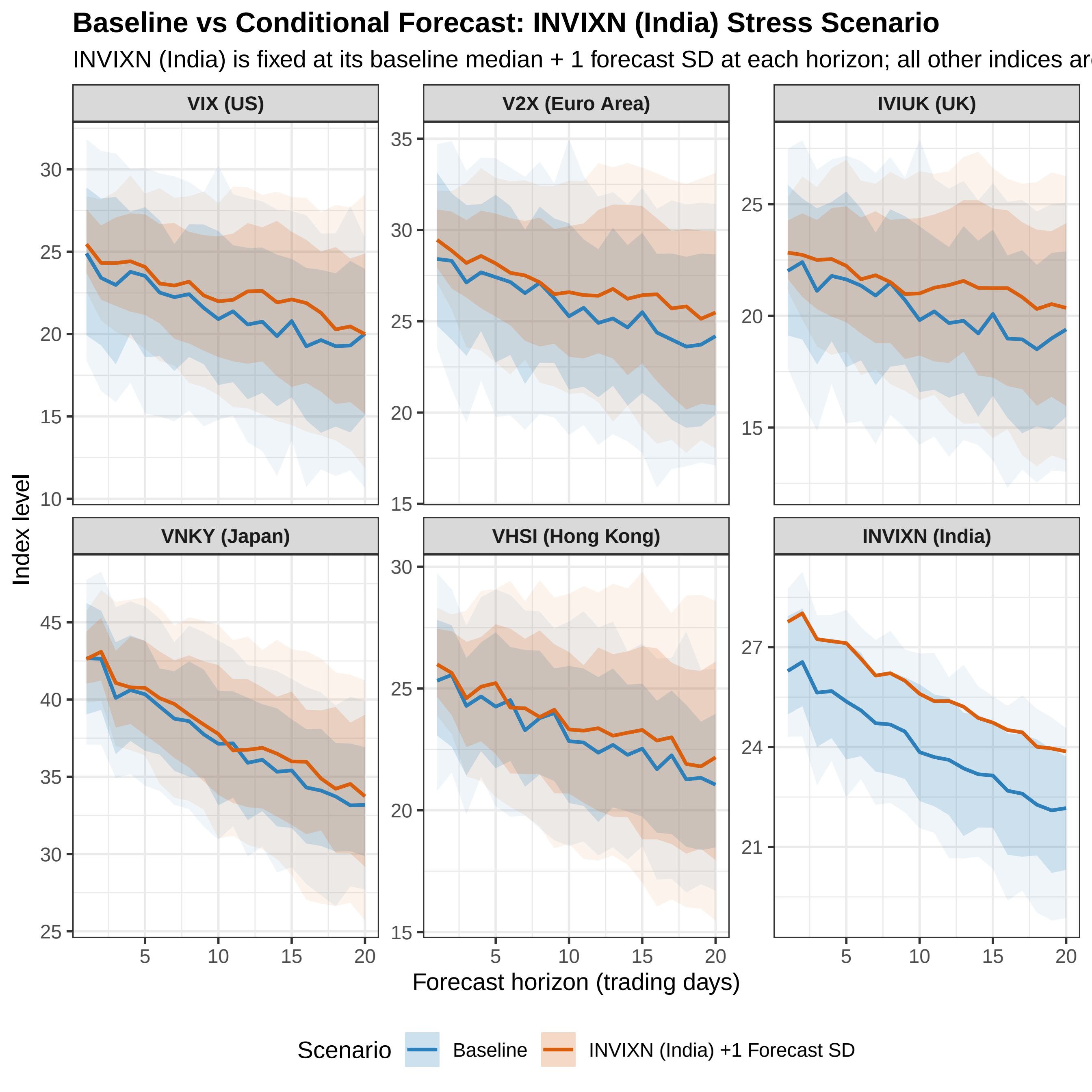

Baseline vs Conditional Forecast: V2X (Euro Area) Stress Scenario
V2X (Euro Area) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
V2X (Euro Area) +1 Forecast SD


Baseline vs Conditional Forecast: IVIUK (UK) Stress Scenario
IVIUK (UK) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are fo
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
IVIUK (UK) +1 Forecast SD

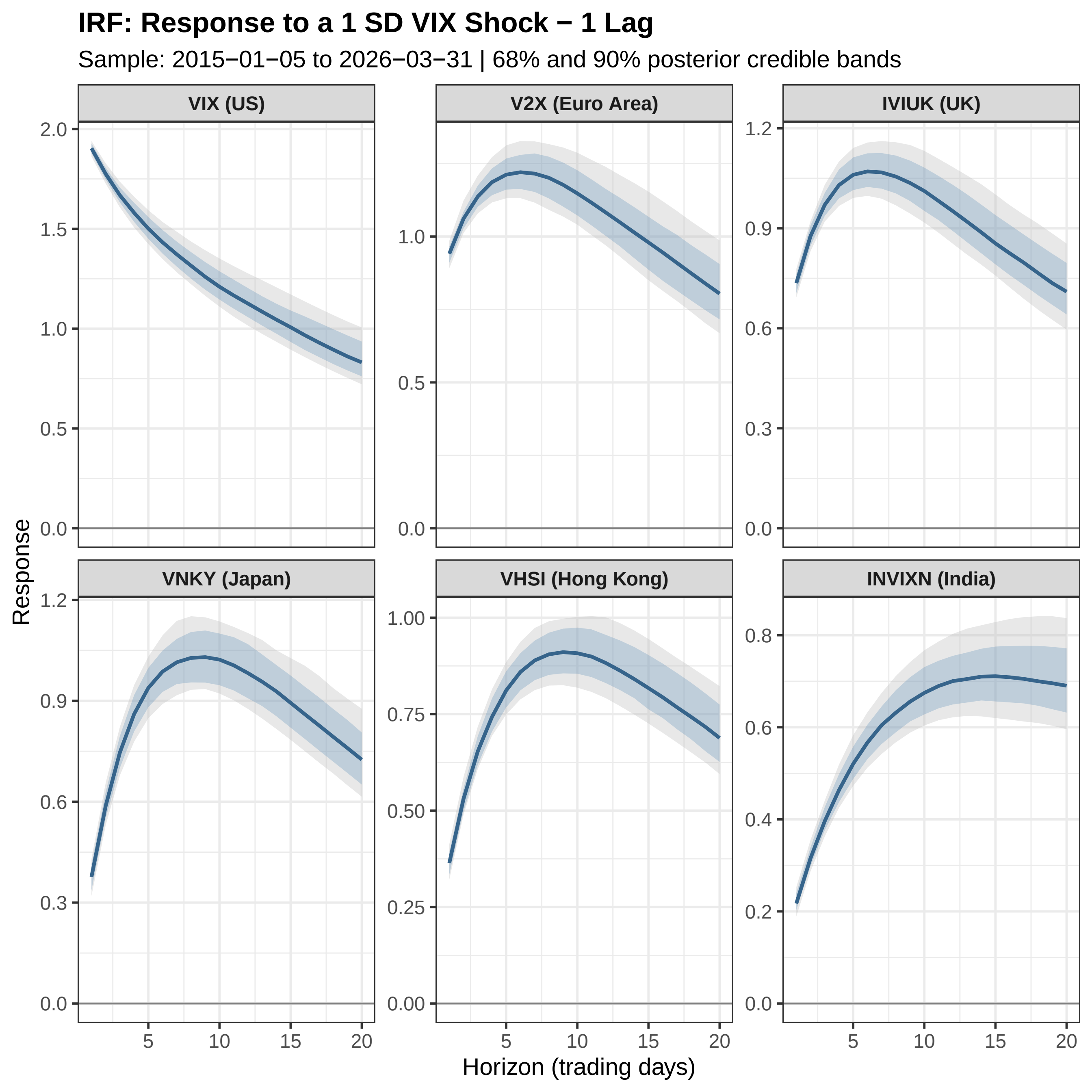


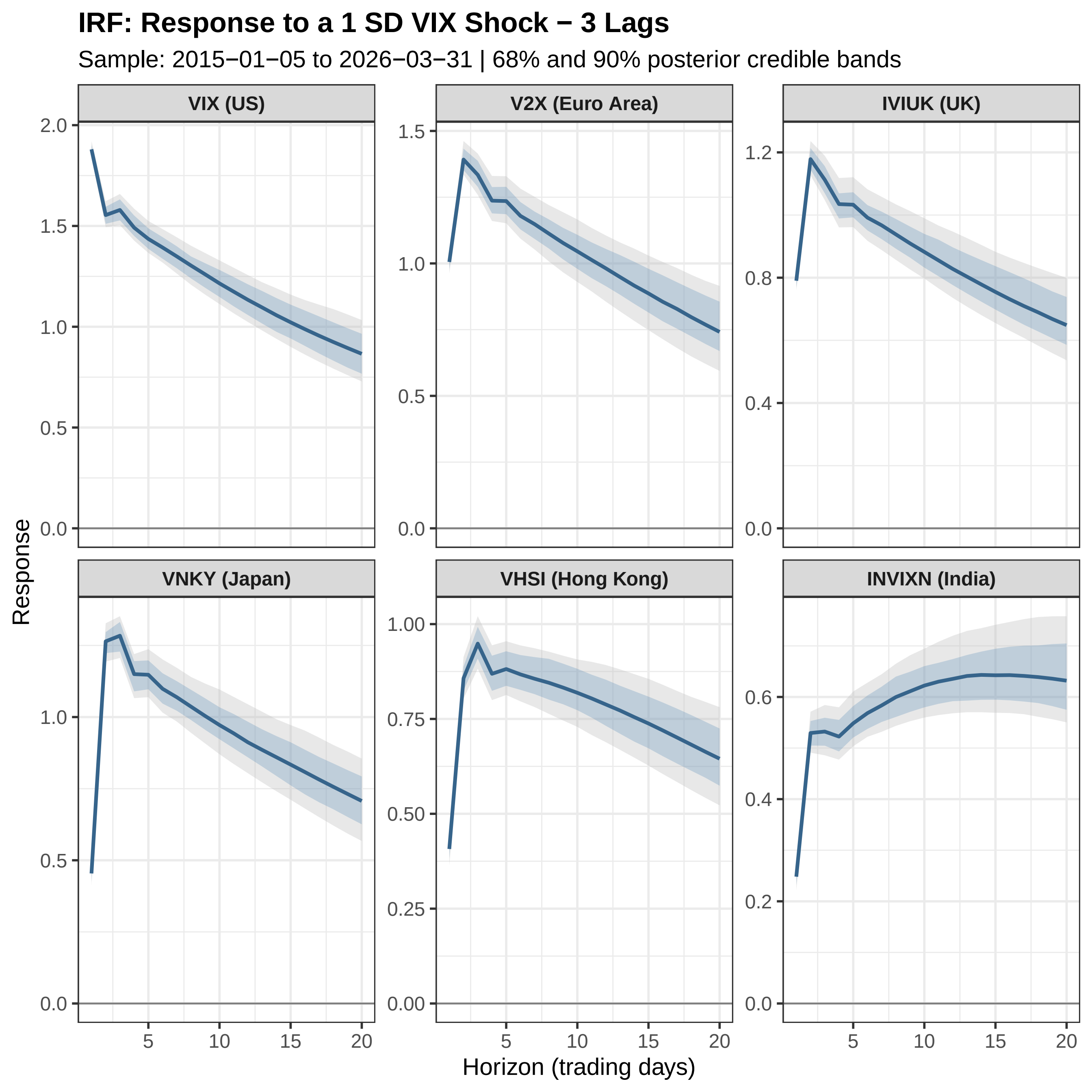
Baseline vs Conditional Forecast: VNKY (Japan) Stress Scenario
VNKY (Japan) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices are
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VNKY (Japan) +1 Forecast SD

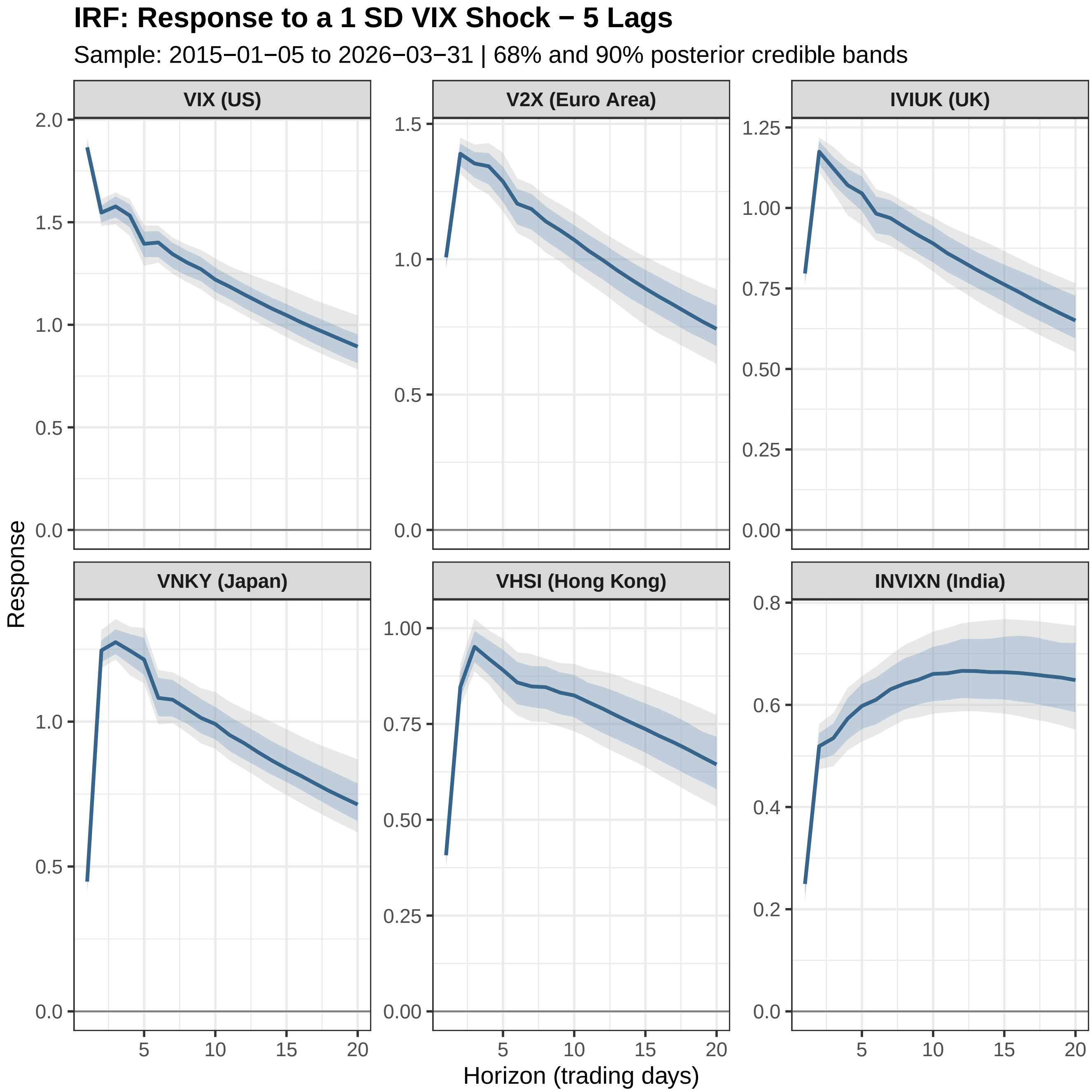

Baseline vs Conditional Forecast: VHSI (Hong Kong) Stress Scenario
VHSI (Hong Kong) is fixed at its baseline median + 1 forecast SD at each horizon; all other indice
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
VHSI (Hong Kong) +1 Forecast SD

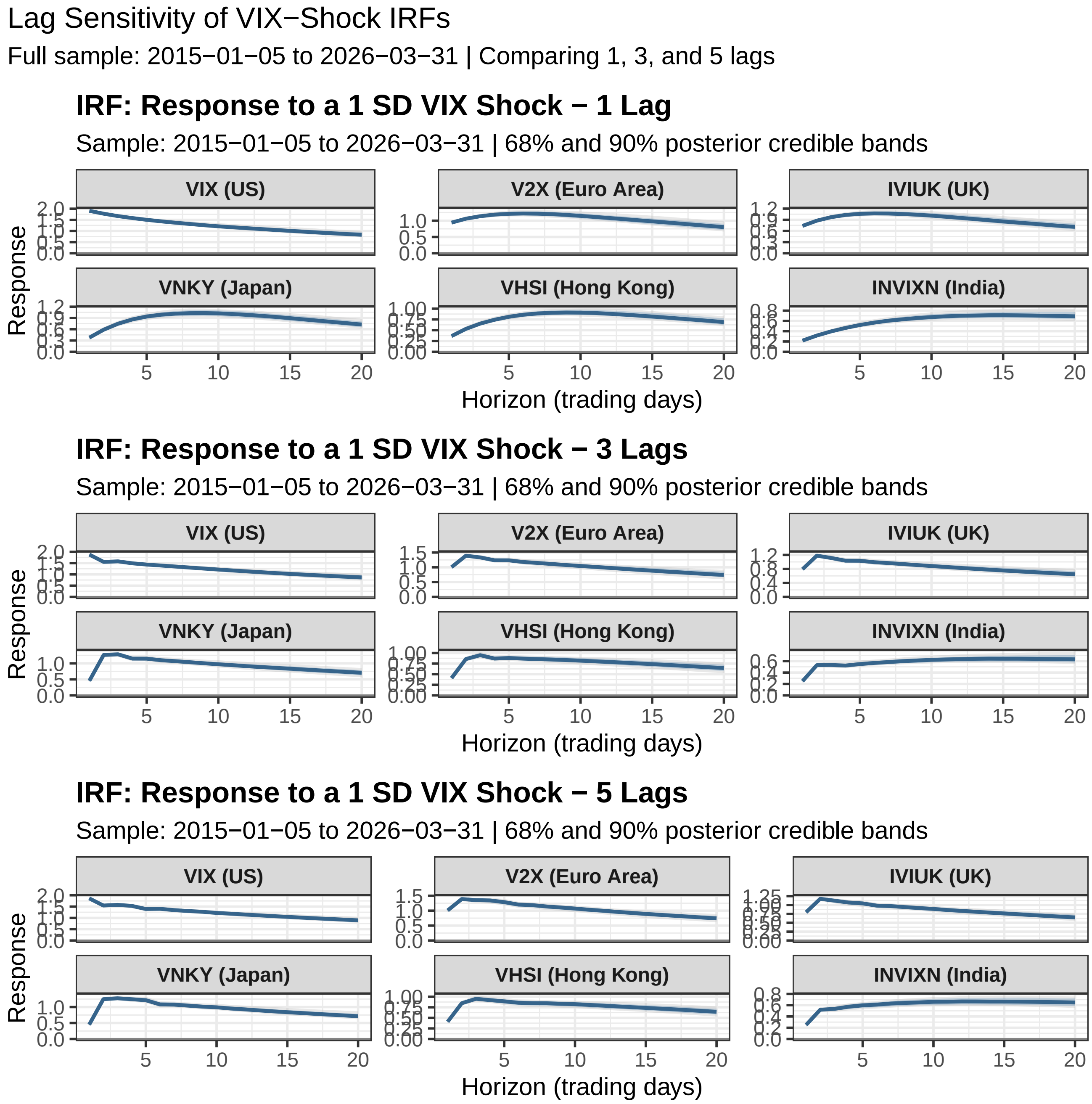


Baseline vs Conditional Forecast: INVIXN (India) Stress Scenario
INVIXN (India) is fixed at its baseline median + 1 forecast SD at each horizon; all other indices ar
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Index level
Forecast horizon (trading days)
Scenario
Baseline
INVIXN (India) +1 Forecast SD


IRF: Response to a 1 SD VIX Shock − 1 Lag
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)


IRF: Response to a 1 SD VIX Shock − 3 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)
0.0
0.5
1.0
1.5
2.0
0.00
0.25
0.50
0.75
1.00
0.2
0.4
0.6
0.8
1.2
5
10
15
20


IRF: Response to a 1 SD VIX Shock − 5 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)


Lag Sensitivity of VIX−Shock IRFs
Full sample: 2015−01−05 to 2026−03−31 | Comparing 1, 3, and 5 lags
IRF: Response to a 1 SD VIX Shock − 1 Lag
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)
IRF: Response to a 1 SD VIX Shock − 3 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)
IRF: Response to a 1 SD VIX Shock − 5 Lags
Sample: 2015−01−05 to 2026−03−31 | 68% and 90% posterior credible bands
VIX (US)
V2X (Euro Area)
IVIUK (UK)
VNKY (Japan)
VHSI (Hong Kong)
INVIXN (India)
Response
Horizon (trading days)